

title

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1 Introduction

1.1 Motivation: The Shifting Landscape of the NBA

Over the last couple of decades, anyone watching the NBA has noticed a big change in how the game is played. We see teams shooting way more three-pointers than before. It makes sense to ask why. In the NBA, the basic rule is that shots made inside the arc are worth 2 points, while shots made further out, behind the arc, are worth 3 points. This simple difference in scoring value seems to have encouraged teams to focus more and more on shooting from distance.

This trend is so noticeable that people often talk about the modern NBA as the "Three-Point Era." Famous players like Stephen Curry really showed how valuable long-range shooting can be, making teams rethink their strategies. But is this just a popular trend, or has the actual importance of making three-pointers for winning games really increased compared to the past? For teams trying to win championships, understanding if and how the value of different shots has changed is really important. In today's world with lots of data available, we felt that using statistical analysis could help quantify this change and maybe offer useful insights for teams trying to optimize how they play.

1.2 Research Question and Approach

So, the main question we wanted to investigate in this project is: How has the *statistical association* between key shooting metrics, especially three-point percentage (3P%), and team success, measured by win rate, changed over the last approximately 15-20 years? Our "story" is essentially testing the common idea that the NBA has truly entered a "three-point era" where making threes is more strongly linked to winning than it used to be.

We need to be careful, though. We are using data collected from watching games (observational data), not from a controlled experiment like an RCT (Randomized Controlled Trial). This means

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we can't easily prove that shooting more threes **causes** more wins, because many other things might be happening at the same time (like better players, different defenses, rule changes). So, our approach is to use statistical tools, mainly multiple linear regression, and specifically interaction terms. These tools allow us to look for strong patterns or **structural evidence** in the data that show if the association between 3P% and winning has really changed significantly over time, while trying to account for some other factors. Our goal is to describe this change accurately based on the statistics, aiming for an interpretation that acknowledges these limitations but points towards the changing dynamics (a quasi-causal goal).

1.3 Thesis Statement

Using data from the 2005-2009 and 2020-2024 NBA regular seasons (source: NBA API), we find the statistical association between three-point percentage (3P%) and win rate differs significantly between these periods: NBA team-seasons (unit of analysis) in the 2020-2024 period with higher 3P% also have statistically significant higher win rates, while NBA team-seasons in the 2005-2009 period did not have statistically significant higher win rates, even after controlling for two-point efficiency and shot selection proportions.

1.4 Report Outline

The remainder of this report details the methodology we used, including data sources, variable definitions, and statistical techniques. We then present the results, including descriptive visualizations and the findings from our regression models. Finally, we conclude by discussing the implications of our results in relation to our thesis statement and acknowledging the limitations of our study.

2 Data Analysis

2.1 Data Preprocess

Our data preprocessing involved three major steps. First, we retrieved the raw data from an API, where each table corresponds to a different season, and each row records a team's performance for that year. Therefore, our unit of analysis is team/year.

As shown in Figure ??, a large portion of the raw variables consisted of rankings (e.g., team rank in various metrics), which we decided to remove, as they are not suitable for our analysis. We also observed that the raw data already included some basic performance statistics, such as win percentage (W_PCT) and three-point shooting percentage (FG3_PCT). However, since the available

statistics were limited to only a few dimensions, we manually calculated additional variables: win rate(WinRate), three-point attempt rate(3P_ratio), three-point shooting percentage(3P%), and two-point shooting percentage(2P%).

We found that the three-point attempt rate and the two-point attempt rate are in a perfect linear relationship (they add up to 1), so we included only the three-point attempt rate in our models to avoid multicollinearity.

Finally, because most of our key variables, such as win rate and shooting percentages, naturally fall within the range $[0, 1]$, we applied normalization to the other numerical variables as well to ensure these variables also fall within the range $[0, 1]$, which maintains the consistency across features.

2.2 Data Visualization

To motivate our investigation into the evolving impact of three-point shooting on team success, we first examine two sets of histograms comparing the 2005-2009 and 2020-2024 NBA regular seasons.

The first set of histograms(Figure 1) shows the three-point shooting percentage (3P%), capturing efficiency on three-point attempts. While the distributions for the two periods overlap significantly, there is a modest rightward shift, indicating that teams have become slightly more accurate from beyond the arc over time. This suggests that the increase in three-point volume has not come at the expense of efficiency, but rather, teams have adapted by developing better shooters and improving shot quality.

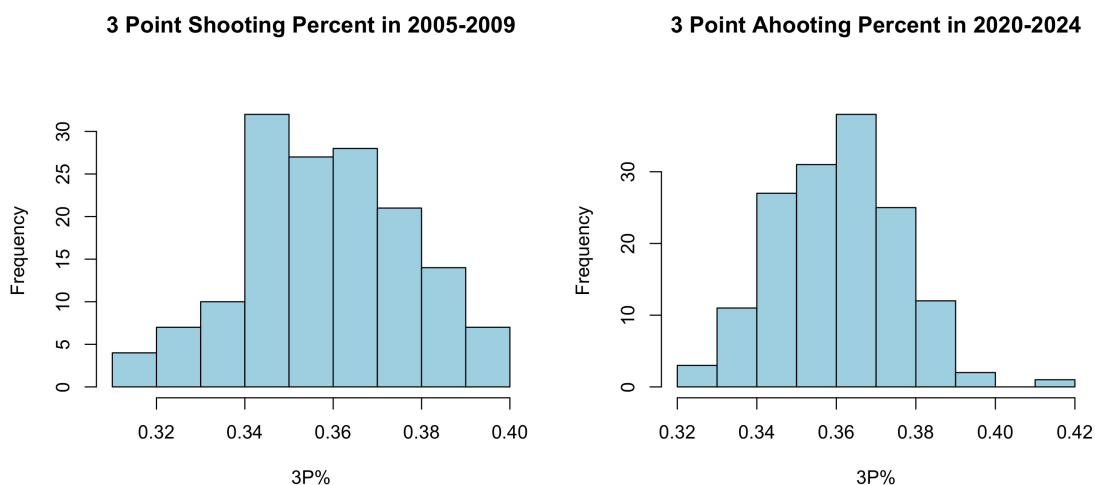


Figure 1: 3 Point Shooting Percent in Two Different Period.

The second set of histograms(Figure 2) displays the three-point attempt rate (3P_ratio), defined as

the proportion of a team's total field goal attempts that are three-pointers. We observe a substantial rightward shift over time: in the 2005–2009 period, teams clustered around a 20% attempt rate, whereas by 2020–2024, the distribution centers around 38–40%. This sharp increase visually confirms the commonly cited notion that the NBA has entered a “small ball” era, characterized by greater reliance on perimeter shooting and faster-paced offenses.

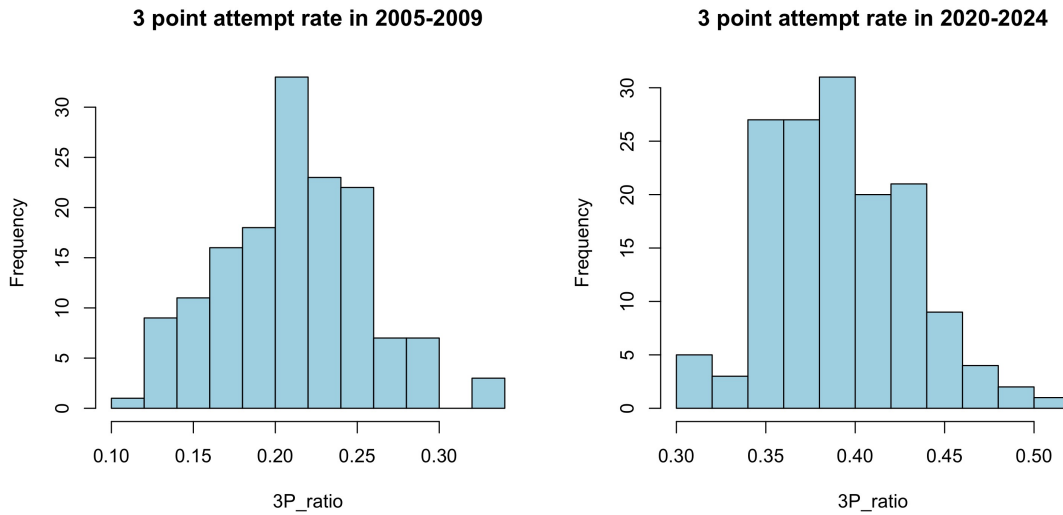


Figure 2: 3 Point Attempt Rate in Two Different Period.

Together, these visualizations provide structural evidence for a fundamental change in league-wide offensive strategy. The sharp increase in three-point attempt rates, coupled with stable or slightly improved three-point accuracy, sets the stage for our central analysis: evaluating whether and how the growing emphasis on three-point shooting translates into higher win rates in the modern NBA.

3 Results

3.1 Initial Analysis: Period-Specific Associations

To get a first look at how relationships might have changed, we fitted separate simple linear regression models for different 5-year periods, specified as follows:

$$\text{WinRate} = \beta_0 + \beta_1 \times \text{3P\%} + \beta_2 \times \text{2P\%} + \beta_3 \times \text{3P_ratio} + \epsilon$$

We focus here on comparing the earliest period (2020–2024). Figure 3 displays the summary outputs for these two models.

```
Call:
lm(formula = WinRate ~ X3P. + X2P. + X3P_ratio, data = mydf_2005_2009)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-0.286612 -0.091560  0.009916  0.080742  0.261863
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -2.0312     0.3044  -6.672 4.87e-10 ***
X3P.          1.8842     0.6407   2.941 0.00381 **
X2P.          3.7431     0.6399   5.850 3.11e-08 ***
X3P_ratio     0.2650     0.2554   1.038 0.30118
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.1216 on 146 degrees of freedom
Multiple R-squared:  0.3694,    Adjusted R-squared:  0.3564
F-statistic: 28.51 on 3 and 146 DF,  p-value: 1.436e-14
```

(a) Model Summary: 2005-2009 Period.

```
Call:
lm(formula = WinRate ~ X3P. + X2P. + X3P_ratio, data = mydf_2020_2024)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-0.282708 -0.060843 -0.004748  0.068760  0.251019
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -2.3217     0.2431  -9.551 < 2e-16 ***
X3P.          4.7683     0.5728   8.324 5.58e-14 ***
X2P.          2.1837     0.4371   4.996 1.65e-06 ***
X3P_ratio    -0.1782     0.2271  -0.784  0.434
```

```
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.1022 on 146 degrees of freedom
Multiple R-squared:  0.4985,    Adjusted R-squared:  0.4882
F-statistic: 48.38 on 3 and 146 DF,  p-value: < 2.2e-16
```

(b) Model Summary: 2020-2024 Period.

Figure 3: Comparison of Parsimonious Linear Regression Model Summaries for Early and Recent Eras.

Comparing the outputs shown in Figure 3, we observe notable differences between the two periods. In the 2005-2009 model (Figure 3a), both two-point percentage (X2P.) and three-point percentage (X3P.) show statistically significant positive associations with WinRate. However, the estimated coefficient for X2P. (approximately 3.74) is considerably larger than that for X3P. (approximately 1.88). Furthermore, the p-value associated with X2P. is much smaller ($p < 0.001$) compared to the p-value for X3P. ($p \approx 0.0038$). In this earlier period, the three-point attempt rate (X3P_ratio) was not found to have a statistically significant association with WinRate.

In contrast, when looking at the 2020-2024 model (Figure 3b), while both X2P. and X3P. remain statistically significant and positively associated with WinRate, their apparent relative importance seems to have reversed. The coefficient for X3P. (approximately 4.77) is now substantially larger than the coefficient for X2P. (approximately 2.18). The very small p-value for X3P. ($p < 0.001$) indicates a very strong statistical association in this recent period. Similar to the earlier period, X3P_ratio remains non-significant. This initial comparison suggests a potential shift: the positive association between three-point efficiency and winning appears stronger in the recent era, while the association for two-point efficiency might have weakened.

To visualize these changes more broadly, we plotted the estimated coefficients and their corresponding p-values (on a reversed log scale) for 'X3P.' and 'X2P.' across four consecutive 5-year periods. Figure 4 presents these trends.

Figure 4a clearly shows the coefficient for 'X3P.' (blue line) steadily increasing over time, while the coefficient for 'X2P.' (red line) generally decreases after the 2010-2014 period. Figure 4b complements this by showing the p-value associated with 'X3P.' decreasing (indicating strengthening statistical significance), while the p-value for 'X2P.', although remaining significant, shows less of a strengthening trend. These descriptive trends strongly indicate a changing relationship and motivate

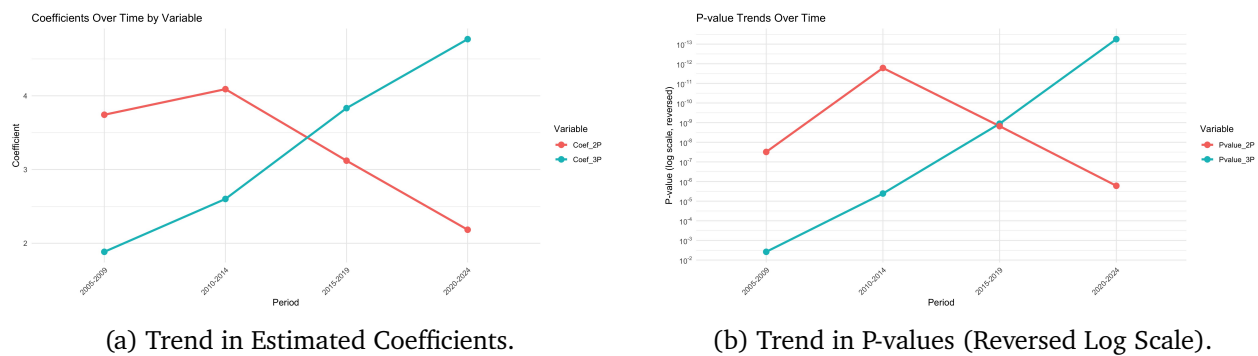


Figure 4: Trends in Coefficients and P-values for 'X3P.' and 'X2P.' from Period-Specific Regressions.

the use of an interaction model to formally test the significance of this observed shift.

3.2 The Interaction Model

To capture the evolving relationship between shooting metrics and team success over time, we fit an interactive linear regression model incorporating both main effects and interaction terms with era. The model is specified as:

$$\begin{aligned} \text{WinRate} = & \beta_0 + \beta_1 3P\% + \beta_2 2P\% + \beta_3 3P_{ratio} + \beta_4 2P_{ratio} + \beta_5 \text{Era} \\ & + \beta_6 (3P\% \times \text{Era}) + \beta_7 (2P\% \times \text{Era}) + \beta_8 (3P_{ratio} \times \text{Era}) \\ & + \beta_9 (2P_{ratio} \times \text{Era}) + \epsilon. \end{aligned}$$

Our model summary is shown in Table 1.

Table 1: Summary of Interaction Model Results (Response: WinRate, based on R output)

Predictor	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.0312	0.2812	-7.224	4.41e-12 ***
X3P.	1.8842	0.5918	3.184	0.001610 **
X2P.	3.7431	0.5910	6.334	9.02e-10 ***
X3P_ratio	0.2650	0.2359	1.123	0.262187
Era (1 = 2020-24)	-0.2905	0.3879	-0.749	0.454448
X3P.:Era	2.8842	0.8641	3.338	0.000954 ***
X2P.:Era	-1.5594	0.7616	-2.047	0.041509 *
X3P_ratio:Era	-0.4431	0.3434	-1.290	0.197976

*Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1*
 Residual standard error: 0.1123 on 292 degrees of freedom
 Multiple R-squared: 0.4301, Adj. R-squared: 0.4165
 F-statistic: 31.49 on 7 and 292 DF, p-value: < 2.2e-16

The results in Table 1 provide the crucial statistical evidence for our study. The key term is the interaction between three-point percentage and era (X3P.:Era). Its coefficient is approximately

2.8842 and is highly statistically significant ($p \approx 0.000954$, '***'). This positive and significant interaction term statistically confirms our central hypothesis: the positive association between a team's X3P. and its WinRate is significantly stronger in the 2020-2024 era (Era=1) compared to the baseline 2005-2009 era (Era=0). Specifically, for every one percentage point increase in X3P., the associated increase in WinRate is estimated to be about 2.88 points higher in the 2020-2024 era than it was in the 2005-2009 era, holding other predictors constant.

Furthermore, the interaction term X2P.:Era has a coefficient of approximately -1.5594 and is statistically significant at the 0.05 level ($p \approx 0.0415$, '*'). This suggests that the positive association between X2P. and WinRate significantly weakened in the later era compared to the baseline era. These findings align perfectly with the descriptive trends shown in Figure 4a. The main effects for X3P. and X2P. represent their associations in the baseline (2005-2009) era, both being positive and significant. Other terms, including X3P_ratio, its interaction with era, and the main effect of 'Era', did not show statistically significant associations in this model. The overall model provides a reasonable fit (Adjusted R-squared ≈ 0.4165) and is highly significant overall (F-statistic p-value $< 2.2e-16$). The strong significance of the X3P.:Era interaction provides the formal statistical validation for our thesis about the changing relationship.

3.3 Model Diagnostics

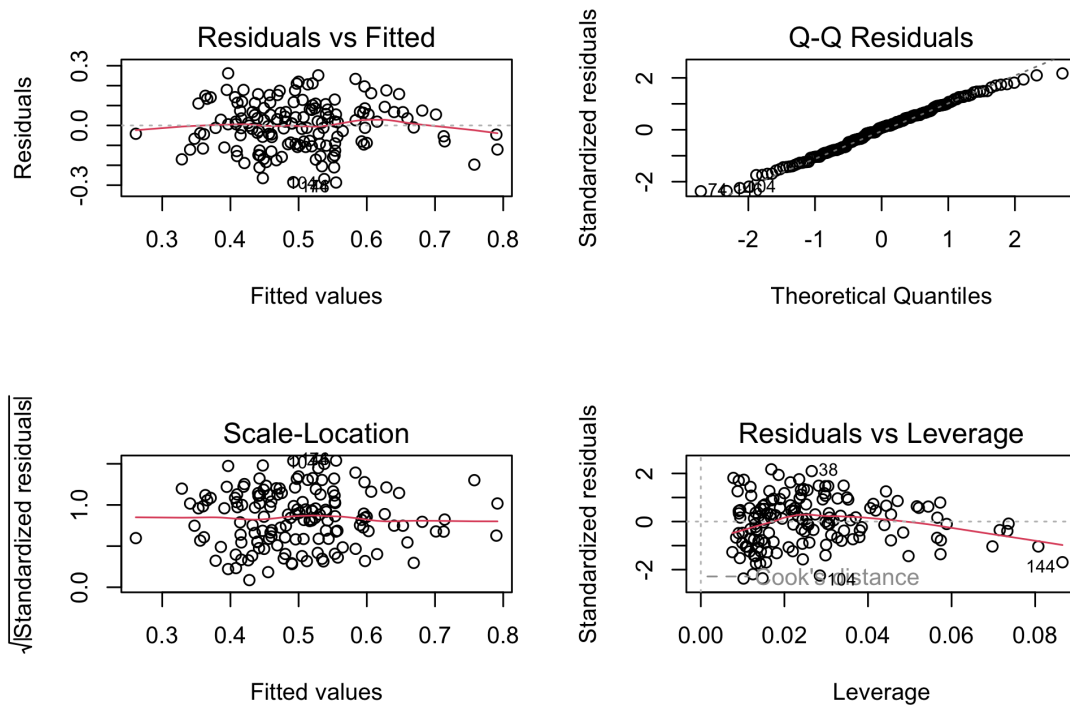


Figure 5: Diagnostic Plots for the Final Interaction Model. (Filename suggests 2005-2009 simple model; verify plot corresponds to interaction model in Table 1).

Finally, we assessed the validity of the assumptions underlying the OLS regression for our final interaction model (results in Table 1). The standard diagnostic plots associated with this model are presented in Figure 5.

Visual inspection of these plots provides confidence in our model. The 'Residuals vs Fitted' plot displays points scattered fairly randomly around the horizontal line at zero, without a clear non-linear pattern or a distinct funnel shape, which supports the assumptions of linearity and constant variance (homoscedasticity). The 'Normal Q-Q' plot shows the standardized residuals falling generally close to the theoretical diagonal line, suggesting that the normality assumption for the residuals is reasonably met, although there might be slight deviations in the tails as is common with real-world data. The 'Scale-Location' plot further examines the homoscedasticity assumption; the points appear relatively evenly spread across the range of fitted values, and the smoothed red line is mostly flat, providing additional support for constant variance. Lastly, the 'Residuals vs Leverage' plot helps identify potentially influential observations. Most points exhibit low leverage. While point 144 shows slightly higher leverage than others, its standardized residual does not appear extreme, and no points seem to significantly exceed typical Cook's distance thresholds (though contours are not explicitly labeled). Therefore, based on this visual assessment, the standard OLS assumptions seem adequately satisfied for our interaction model, lending validity to the statistical inferences we have drawn from

it.

4 Conclusion

4.1 Summary of Findings in Relation to Thesis

This project statistically investigated the perceived shift towards a "Three-Point Era" in the NBA by examining the changing association between shooting metrics and team success over two decades. Our findings support our main thesis: Using data from the 2005-2009 and 2020-2024 NBA regular seasons (source: NBA API), we find the statistical association between three-point percentage (3P%) and win rate differs significantly between these periods: NBA team-seasons in the 2020-2024 period with higher 3P% also have statistically significant higher win rates, while NBA team-seasons in the 2005-2009 period did not have statistically significant higher win rates, even after controlling for two-point efficiency and shot selection proportions. The key evidence for this conclusion is the statistically significant interaction term between three-point percentage and era (Era:X3P.) found in our regression analysis (Table 1).

4.2 Interpretation and Implications

The significant interaction term quantifies the degree to which the statistical relationship between three-point shooting efficiency and winning has strengthened. Our results estimate that the positive association between 3P% and WinRate was substantially stronger (by approximately 2.88 units based on the coefficient) in the 2020-2024 period compared to 2005-2009. This provides strong quantitative backing to the narrative of a "Three-Point Era," showing a clear structural shift in league dynamics where efficient scoring from behind the arc is much more closely linked to team success than it was previously. While our observational approach cannot definitively prove causation, the magnitude and significance of this *change* in association is consistent with the idea that the underlying strategic value of the three-point shot has indeed increased over time.

4.3 Limitations of the Analysis

Several limitations must be considered when interpreting these results. Primarily, as this study relies on observational data, we cannot make definitive causal claims. The observed associations, and the changes in them, could be influenced by numerous unobserved confounding variables that evolved alongside shooting strategies. Potential confounders include changes in defensive rules and strategies over the years, the overall evolution of player skill sets (especially the rise of prolific shoot-

ers), influential coaching philosophies adapting to or driving trends, the impact of specific superstar players on their teams' style and success, uncaptured variations in team defense quality, or even factors like player injuries or strength of schedule within specific seasons. While our model controls for key efficiency and shot selection metrics, it cannot account for all such complex, potentially time-varying confounders. Other limitations include the specific choice of time periods for comparison and the inherent assumptions of the linear regression model framework.

4.4 Future Research

Future research could aim to address these limitations by incorporating data on defensive metrics, coaching changes, or player-level statistics if available. Exploring different time frames or using panel data methods might also provide further insights. Applying more advanced methods aimed at causal inference from observational data could be valuable, though potentially challenging.

4.5 Concluding Remarks

In conclusion, this study provides robust statistical evidence, highlighted by a significant interaction effect, documenting the substantially stronger association between three-point efficiency and team success in the modern NBA compared to the mid-2000s.